RAJALAKSHMI ENGINEERING COLLEGE

Hasini N 250701237-2025-CSE

V2

Started on	Friday, 14 August 2026, 3:33 PM
State	Finished
Completed on	Friday, 14 August 2026, 3:38 PM
Time taken	4 mins 50 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

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Falling back to raw text area.

```
#include<stdio.h>
int main(){
int n;
scanf("%d", &n);
int i = 1;
int a = 1;
int count = 3;
while(a<=n){
count++;
i++;
count++;
a+=i;
count++;
}
printf("%d", count);
return 0;
}
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Hasini N 250701237-2025-CSE

V2

Started on	Friday, 14 August 2026, 3:38 PM
State	Finished
Completed on	Friday, 14 August 2026, 3:40 PM
Time taken	1 min 48 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
    int n;
    int count = 0;
    scanf("%d", &n);
    if(n==1){
        count = 1;
    }
    else{
        count = 5*n+2;
    }
    printf("%d", count);
    return 0;
}
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔

	Input	Expected	Got	
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Hasini N 250701237-2025-CSE

V2

Started on	Friday, 14 August 2026, 3:40 PM
State	Finished
Completed on	Friday, 14 August 2026, 4:10 PM
Time taken	29 mins 10 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

void Factor(int num) {
    int count = 0;
    int i;
    for (i = 1; i <= num; ++i) {
        count++;
        count++;
        if (num % i == 0) {
            count++;
        }
    }
    count++;
    printf("%d", count);
}
int main() {
    int n;
    if (scanf("%d", &n) == 1) {
        Factor(n);
    }
    return 0;
}
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Hasini N 250701237-2025-CSE

V2

Started on	Friday, 21 August 2026, 3:05 PM
State	Finished
Completed on	Friday, 21 August 2026, 3:10 PM
Time taken	4 mins 21 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include<stdio.h>
int main(){
    int n,c=0;
    scanf("%d", &n);
    c++;
    for(int i=n/2; ;i++){
        c++;
        if(i>=n){
            break;
        }
        for(int j=1; ;j=2*j){
            c++;
            if(j>=n){
                break;
            }
            for(int k=1; ; k = k*2){
                c++;
                if(k>=n){
                    break;
                }
                c++;
            }
        }
    }
    printf("%d", c);
    return 0;
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Hasini N 250701237-2025-CSE

V2

Started on	Friday, 21 August 2026, 3:10 PM
State	Finished
Completed on	Friday, 21 August 2026, 3:13 PM
Time taken	3 mins 7 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include<stdio.h>
int main(){
    int n, rev=0, rem;
    int c=0;
    scanf("%d", &n);
    c++;
    while(n!=0){
        c++;
        rem = n%10;
        c++;
        rev = rev*10+rem;
        c++;
        n/=10;
        c++;
    }
    c++;
    c++;
    printf("%d", c);
    return 0;
}
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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